

Eclampsia Case



Section I: Scenario Demographics

Scenario Title:	Eclampsia		
Date of Development:	06/04/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
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Section III: Curriculum Integration

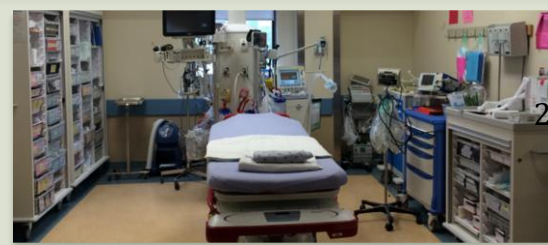
Learning Goals & Objectives	
Educational Goal:	To practice the management of eclampsia, a relatively rare presentation with important critical care steps in its management.
CRM Objectives:	Communicate effectively with team and consultants to provide expedient patient care.
Medical Objectives:	1) Recognize potential pre-eclampsia/HELLP and initiate appropriate diagnostic workup. 2) Administer appropriate dose of magnesium sulfate for eclampsia, including repeated doses. 3) Recognize apnea as a possible consequence of magnesium administration and appropriately work through differential diagnosis and management of apnea.

Case Summary: Brief Summary of Case Progression and Major Events
30 year-old female, G1P0 at 32 weeks, presents to the ED with headache, blurred vision, nausea, and vomiting. Her arrival BP is 165/115. As the team coordinates her initial workup, the patient will begin to seize. She will not stop seizing until magnesium sulfate is given. The patient will then require intubation for respiratory depression. The case will end post intubation when the patient has been referred to OB.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.
Magee LA, Pels A, Helewa H, et al. SOGC Clinical Practice Guideline: Diagnosis, evaluation and management of the hypertensive disorders of pregnancy: executive summary. J Obstet Gynaecol Can 2014;36(5):416-438



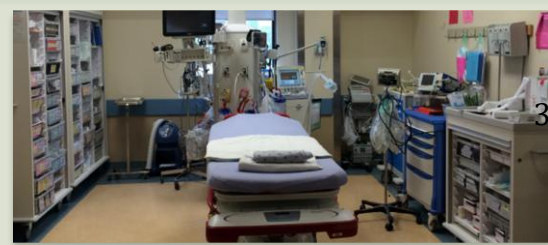
Eclampsia Case



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
<i>Select most important dimension(s)</i>			
Confederates	Brief Description of Role		
Nurse	Simulates seizures when appropriate both by shaking bed and by stating "I think the patient is seizing."		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Pillow/clothes to mimic appearance of pregnant patient.			
Appropriate adjuncts to help simulate seizure.			
E. Approximate Timing			
Set-Up:	5 min	Scenario:	15 min
		Debriefing:	20 min

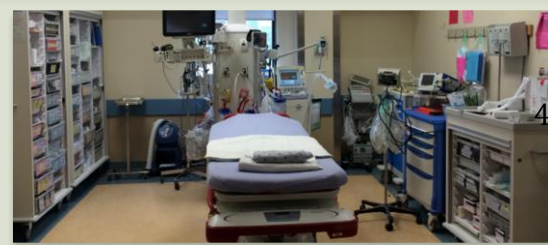
Eclampsia Case



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case			
Miranda Hamm presents to your local tertiary care ED complaining of a headache. She is a 30 year old G1P0 at 32 weeks. She has had a headache since last night. This morning she started feeling nauseous and began vomiting. Now her vision feels blurred, so she came for assessment.			
B. Patient Profile and History			
Patient Name: Miranda Hamm		Age: 30	Weight: 75kg
Gender: ☐ M ☒ F		Code Status: Full	
Chief Complaint: headache			
History of Presenting Illness: Headache since last night. Circumferential and pounding. Thought it was her normal migraine but was afraid to take any medication because of pregnancy. Then started having nausea and vomiting. Didn't really sleep all night. Now headache is worse and vision feels blurry. She's never had blurred vision before.			
Past Medical History:	Migraines	Medications:	None.
Allergies: None.			
Social History: Lives with husband.			
Family History: No relevant disorders.			
Review of Systems:	CNS:	Headache since last night, circumferential and pounding.	
	HEENT:	Blurred vision this morning. "Feels like I'm looking through fog."	
	CVS:	Nil.	
	RESP:	Nil.	
	GI:	Nauseous all day yesterday. Vomiting last night and this morning.	
	GU:	Nil.	
MSK:	Nil.	INT:	Nil.
C. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☒ Monitor On, no data displayed	
☐ Monitor on Standard Display			
HR: 110/min	BP: 175/115	RR: 20/min	O ₂ SAT: 96% RA
Rhythm: Sinus tach	T: 36.5°C	Glucose: 6.4 mmol/L	GCS: 15 (E4 V5 M6)
General Status: Well-looking, but appears in pain.			
CNS:	A+Ox3. No facial asymmetry. No focal deficits. Reflexes 2+.		
HEENT:	PERLA 3mm. No papilledema.		
CVS:	Normal S1/S2. No murmur.		
RESP:	In no distress. Appropriately tachypneic for pregnancy.		
ABDO:	Gravid, non-tender uterus. Some RUQ tenderness. If FHR checked, it's 160.		
GU:	Nil.		
MSK:	Nil.	SKIN:	Nil.

Eclampsia Case



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 110/min BP: 175/115 RR: 20/min O ₂ SAT: 96% T: 36.5°C	A+Ox3. Answering questions. Slightly tachypneic. RUQ tenderness.	<u>Learner Actions</u> ☐ IV, monitors ☐ Assess FHR (160) ☐ Blood work and urinalysis ☐ Hx and Px	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 3 minutes → 2. Seizure
2. Seizure HR → 130	Actively seizing. Nurse to cue learners (I think she's seizing!) and initiate seizure.	<u>Learner Actions</u> ☐ Administer MgSO ₄ 4g iv bolus then 1-2g/hr iv ☐ Check cap sugar (6.4) ☐ Call OB/GYN for delivery	<u>Modifiers</u> - Benzo given → no change to seizures <u>Triggers</u> - MgSO ₄ given → 3. Apnea
3. Apnea HR → 110 RR → 6 → 0 O ₂ SAT → 90 → 88% BP → 165/115	Not seizing. RR slowly decreasing with associated desaturations. Not responsive.	<u>Learner Actions</u> ☐ Initiate BVM ☐ Stop MgSO ₄ infusion ☐ Give Ca gluconate 1g iv ☐ Administer induction ☐ Administer paralytic	<u>Modifiers</u> - BVM initiated → O ₂ SAT 93% - Propofol or midazolam for induction → BP 160/110 - Calcium gluconate → no effect <u>Triggers</u> - Intubated → 4. Post-intubation
4. Post-intubation HR → 110 RR → 12 (vent) O ₂ SAT → 98% BP → 165/115	Intubated and sedated.	<u>Learner Actions</u> ☐ Call OB/GYN ☐ Call ICU ☐ Post-intubation CXR ☐ Restart MgSO ₄ infusion ☐ Labetolol or hydralazine ☐ Consider CT head	<u>Modifiers</u> - Antihypertensive → BP 145/100 <u>Triggers</u> - Antihypertensive given → 5. Resolution - 15 min → 5. Resolution
5. Resolution BP → 145/100	Intubated and sedated.		OB/GYN arrives to take patient for C/S. → Team to explain case to consultant. End scenario.

Eclampsia Case



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What is the dosing for magnesium sulfate in eclampsia? In pre-eclampsia? 2) When should you consider an anti-hypertensive in eclampsia? 3) Why did this patient become apneic? What are the other associated side effects of magnesium? What is a possible treatment to reverse these effects?	
Key Moments	
Recognition of possible pre-eclampsia HELLP syndrome and initiation of key blood work.	
Administering MgSO4 to treat eclampsia.	
Recognizing apnea as side effect of MgSO4.	